Transfer: circuits and induction

antocciphysics | TEACHER KEY | CPA

Learning target: Apply circuit relationships and explain an induction observation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 15; 3 course days later.*
A generator. Some energy can become heating, friction or sound.

02. Explain a worked example (worked)
Ammeter in series measures 2 A; voltmeter across resistor measures 12 V; input power 24 W.

03. Look and explain (model)
Both resistor ends share the battery’s two nodes.

04. Build the solution (practice)
2 A and 1 A; total 3 A; 18 W.

05. Use the idea (reason)
Motion changes flux, inducing emf; mechanical work supplies energy when a current transfers energy to the circuit.

06. Choose a method (mixed) *Review: Unit 7, Day 6; 12 course days later.*
V=IR=24 V.

07. Exit check (exit)
30 J; heat or sound.

08. STRETCH: Apply it another way (stretch)
Equivalent 6 ohms; same 2 A total and 24 W; each branch 1 A and 12 W.

Teaching note: Daily transfer practice. Keep any separately assigned formal assessment independent.

Transfer: circuits and induction

antocciphysics | TEACHER KEY | HONORS

Learning target: Apply circuit relationships and explain an induction observation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 15; 3 course days later.*
Combined fraction=0.4402 or 44.02%, multiplying the two efficiencies.

02. Explain a worked example (worked)
Ammeter in series measures 2 A; voltmeter across resistor measures 12 V; input power 24 W.

03. Represent and interpret (model)
Both resistor ends share the battery's two nodes.

04. Solve and justify (practice)
$R_{parallel}$=4 Ω, total 8 Ω; 1.5 A total; .75 A per branch; 18 W.

05. Analyze the reasoning (reason)
Motion changes flux, inducing emf; mechanical work supplies energy when a current transfers energy to the circuit.

06. Choose a method (mixed) *Review: Unit 7, Day 6; 12 course days later.*
R=40 Ω; V/I is constant at these points under the measured conditions.

07. Exit check (exit)
1 V.

08. STRETCH: Apply it another way (stretch)
Total resistance 4 ohms, current 3 A, total 36 W. Series 18 W; branches 9 W each; sum 36 W.

Teaching note: Daily transfer practice. Keep any separately assigned formal assessment independent.